

Valuation Practice and Theory: Traditional Models and a Jump-Diffusion Framework

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Abstract

This study evaluates the empirical performance of canonical corporate valuation techniques and develops a jump-diffusion, real-options framework for valuing illiquid venture investments. We apply Discounted Cash Flow (DCF), Asset-Based Valuation (ABV), and Comparable Company Analysis (CCA) to a sample of over 250 firms engaged in M&A and IPO transactions. ABV systematically understates enterprise value and is useful, at most, as a lower-bound check. Although theoretically well-grounded and widely adopted, the DCF exhibits pronounced input sensitivity, resulting in a wide dispersion of estimates under reasonable parameter variations. CCA results are likewise heterogeneous; however, EV/Sales emerges as the comparatively more robust multiple—particularly in cyclical industries—relative to earnings-based ratios. In the second part, we develop and test a jump-diffusion real-options model on over 3,400 venture capital-financed startup investments. By explicitly accommodating discontinuities, embedded managerial flexibility, and market illiquidity, the model aligns most closely with observed transaction values when the investment size is approximately 25–35% of pre-money valuation; reliability deteriorates for larger stakes (>35%) and effectively collapses beyond 75%, reflecting intrinsic boundary conditions of the framework. Collectively, the findings delineate where traditional methods remain informative, identify their practical limitations, and articulate conditions under which a real-options, jump-diffusion approach is warranted, contributing to a more operational valuation toolkit spanning public, private, and illiquid venture contexts.

Keywords: Corporate valuation; DCF; Asset-based valuation; Comparable multiples; Real options; Jump-diffusion; M&A; IPO; Private Equity; Venture capital

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1 Introduction

Valuation lies at the heart of corporate finance. As technological shifts, policy regimes, and liquidity conditions evolve, the choice and implementation of valuation methods increasingly determine how credibly firm value is inferred and how capital is allocated across sectors and regions. Recent research has renewed its focus on the reliability of these methods so that estimated value better reflects firms’ risk–return prospects amid financial, strategic, operational, and regulatory uncertainties.

A prominent frontier involves illiquid assets—most notably venture-backed firms—where daily prices are unobservable, comparables are sparse, and payoff distributions are markedly asymmetric. Early-stage startups crystallize these frictions: limited financial histories, profitability that is either nonexistent or highly uncertain, and discontinuous growth dynamics that alternate between rapid scaling and abrupt failure. In such settings, classical techniques premised on rich accounting trails or stable market peers can yield fragile signals unless they explicitly accommodate optionality, abandonment, and time-to-market.

At the same time, public and private transaction markets—Mergers & Acquisitions (M&A) and Initial Public Offerings (IPOs)—provide frequent, consequential benchmarks for value. Acquisitions reconfigure capabilities, scale, and competitive positioning; listings open access to public capital and can reset the cost of funds. Yet both arenas are vulnerable to systematic under- or overvaluation when techniques are misapplied, with repercussions for deal terms, dilution, and post-transaction performance. Advancing valuation practice, therefore, bears directly on corporate policy, investor selection, and intermediary incentives.

This study contributes along two dimensions. First, it reassesses the empirical performance of canonical techniques—Discounted Cash Flow (DCF), Asset-Based Valuation (ABV), and Comparable Company Analysis (CCA)—against realized deal values. The DCF approach projects and discounts firm-specific cash flows using parameters that reflect business and industry risk, offering an intrinsically forward-looking measure of value Damodaran (2023) and Koller et al. (2020). In contrast, ABV anchors valuation to the most recent balance-sheet configuration—assets net of liabilities—providing a static snapshot that abstracts from growth options and operating trajectories Damodaran (2006) and Penman (2013). CCA, finally, estimates value by mapping a focal firm to a peer set and applying market multiples that embody prevailing conditions across size, sector, and operating structure Liu et al. (2002) and Olbert (2024). Each method is widely used, yet each can generate dispersion or bias when inputs, accounting quality, or peer definitions are strained.

Second, to address illiquidity and discontinuity directly, the thesis develops and tests a real-options, jump-diffusion framework in which firm value evolves as a continuous process punctuated by rare positive or negative jumps that capture transformational successes or collapses; Monte Carlo simulation traces path ensembles relative to an investment-size barrier, translating managerial flexibility and tail risk into valuation metrics Gibson et al. (2022).

The empirical design aligns methods to the environments where they are admissible. For data-mature firms engaged in transactions, DCF, ABV, and CCA are implemented on a large cross-section of companies with observed M&A and IPO outcomes from *Orbis BVD*. For early-stage, illiquid investments, the jump-diffusion model is estimated and validated on venture-capital deals sourced from *Eikon Refinitiv*. Across both settings, inputs are standardized, application protocols are documented, and accuracy is evaluated against realized values, with heterogeneity profiled by sector, geography, and deal type. Robustness checks consider input perturbations, alternative pricing windows, and peer-set constructions; for the option-based model, diagnostics address parameter identifiability, path dependence, and barrier calibration.

The analysis is organized around two guiding questions:

1. How accurately do traditional approaches (DCF, ABV, CCA) recover transaction values in M&A and IPO datasets, and under what conditions do errors concentrate?
2. Where traditional methods are inapplicable—most notably for illiquid, early-stage assets—does a jump-diffusion, real-options specification provide an empirically valid and decision-useful alternative?

The remainder of the thesis proceeds as follows. Section 1 surveys the valuation literature, clarifying what established models capture—and systematically miss—when mapped to contemporary deal environments. Section 2 codifies implementation details for DCF, ABV, CCA, and the jump-diffusion framework, harmonizing dispersed guidance into reproducible procedures. Sections 3 and 4 report the empirical results: traditional models benchmarked to *Orbis BVD* M&A/IPO values, and the real-options jump-diffusion applied to *Eikon Refinitiv* venture investments, with comparative accuracy and error-structure diagnostics. Section 5 concludes by summarizing findings, acknowledging limitations, and outlining extensions, including incorporating market-implied information, learning about jump intensities across financing rounds, and stress-testing valuations under alternative policy and rate regimes.

2 Literature Review

This chapter reviews the principal approaches to business valuation, drawing on widely cited academic contributions and practitioner manuals to assemble a coherent, up-to-date map of the field. The discussion proceeds methodically through four pillars. First, we outline the DCF framework and the conditions under which its outputs are decision-useful. Second, we examine asset-based approaches that anchor value in accounting stocks rather than prospective flows. Third, we survey market-multiple methods (Comparable Company Analysis, CCA) and the sectoral contexts in which particular ratios dominate. Finally, we present jump-diffusion, real-options-based models as a recent and more sophisticated response to illiquidity, discontinuities, and timing flexibility in investment.

The DCF method discounts firm-specific expected cash flows—including a terminal value—at a rate intended to capture systematic and industry risk, thereby yielding an intrinsic value grounded in forward-looking fundamentals. Its appeal lies in global standardization, reliance on tangible financial data, and versatility in settings where cash flows are reasonably forecastable. Yet the same mechanics expose substantive fragilities: estimates are highly sensitive to growth and discount-rate assumptions, they abstract from the flexibility to defer or abandon (i.e., the timing option), and they struggle precisely where cash flows are uncertain or absent—conditions common in illiquid markets Gibson et al. (2022). In practice, the quality of DCF output turns on careful construction of both the discount rate and the terminal value; mis-specification of WACC or perpetuity growth propagates disproportionately into valuation error Damodaran (2019) and Koller et al. (2020).

Asset-Based Valuation (ABV) instead treats the firm as a balance-sheet aggregation of assets and liabilities, with equity measured as their difference. This renders ABV especially suited to liquidation contexts, distress, or situations where a conservative floor is preferable to an aspirational forecast Damodaran (2006) and Palepu et al. (2022). The trade-off is clear: by construction, ABV de-emphasizes (or omits) going-concern earning power Damodaran (2009) and inherits the distortions of historical cost accounting, which may depart from current economic value Penman (2013). Consequently, ABV is best employed as a complement—e.g., a lower-bound check—rather than a stand-alone estimator when growth options are salient.

Comparable Company Analysis (CCA) infers value by mapping a target to a peer set and transferring

market-implied ratios. Its strengths are simplicity, alignment with prevailing pricing, and broad applicability to public and private firms. Selection of the multiple is context-dependent. The Price/Earnings (P/E) ratio is widely preferred across many GICS sectors for its connection to earnings expectations Olbert (2024), but becomes unreliable with negative earnings Demirakos et al. (2004). Financials often pivot to Price/Book (P/B), where book equity better tracks economic value Nissim (2012). Price/Cash Flow (P/CF) can be informative in energy and IT/hardware where cash generation is central Olbert (2024). On an enterprise basis, EV/EBITDA is common in energy and telecommunications Huikku and Pöyhä (2020) and Glaum and Friedrich (2006), EV/EBIT gains relevance when depreciation materially affects operating economics, and EV/Sales is used in high-revenue settings though it ignores profitability Olbert (2024). Despite ubiquity, CCA carries methodological pitfalls—circularity in peer selection and survivorship bias—that can induce systematic skew if left unmanaged.

For illiquid assets and early-stage ventures, jump-diffusion models from the real-options tradition offer a more faithful representation of risk and managerial flexibility. These models treat value as a stochastic process with a continuous diffusion component and discrete jumps, capturing transformative successes or abrupt failures. They explicitly value the option to wait for favorable states and provide a richer risk management lens than static DCF. Foundational work formalized sequential investment options Majd and Pindyck (1987) and techniques for processes with jumps Weron et al. (2004). Building on this base, Gibson et al. (2022) develop and implement a jump-diffusion, real-options specification for venture capital valuation, demonstrating advantages in quantifying both value and the timing of commitment. In environments characterized by volatility, sparse comparables, and discontinuities—precisely where classical methods falter—the jump-diffusion framework brings market breaks, option value, and timing uncertainty into the core of the valuation exercise.

3 Empirical models

This chapter turns from concepts to execution, detailing how each valuation approach can be implemented through clear, reproducible procedures. The objective is to guide the reader across the operational steps that translate the literature review into practice, offering a unified view of the end-to-end valuation workflow. The exposition is organized into four sections: the first implements the Discounted Cash Flow (DCF) methodology; the second covers asset-based approaches; the third sets out the market-multiples procedure; the fourth operationalizes the jump-diffusion, real-options framework.

3.1 DCF

This section implements the Discounted Cash Flow (DCF) method in four steps, following standard references such as Damodaran (2023) and Koller et al. (2020).

Step 1: Forecast free cash flow Free Cash Flow (FCF) is the cash available to both debt and equity holders after funding operations and reinvestment needs damodaran2023. Over an explicit forecast horizon,

$$\text{FCF} = \text{EBIT} - \text{Taxes} + \text{D}$$

with the usual components:

$$\begin{aligned}
\text{EBIT} &= \text{Earnings Before Interest and Taxes} \\
\text{Taxes} &= \text{Tax expense based on EBIT} \\
\text{D\&A} &= \text{Depreciation and Amortization} \\
\text{CapEx} &= \text{Capital Expenditures} \\
\Delta\text{NWC} &= \text{Change in Non-Cash Working Capital}
\end{aligned}$$

The detailed forecasting conventions (margins, reinvestment rates, working-capital turns) are reported in Appendix A.1.

Step 2: Compute the WACC Because projected cash flows arrive in future periods, they must be discounted to present value at a rate consistent with the firm’s opportunity cost of capital. The Weighted Average Cost of Capital (WACC) aggregates the cost of equity and the after-tax cost of debt in proportion to market value weights berk2020,koller2020:

$$\text{WACC} = \frac{E}{E+D} R_E + \frac{D}{E+D} R_D (1-T), \quad (2)$$

where E is equity, D is debt, R_E the cost of equity, R_D the cost of debt, and T the marginal tax rate. The cost of equity is often obtained via CAPM berk2020:

$$R_E = R_f + \beta_i (E[R_m] - R_f), \quad (3)$$

with R_f the risk-free rate, β_i the levered equity beta, and $E[R_m] - R_f$ the equity risk premium.

Parameter sourcing Inputs are drawn from standard practitioner and policy sources used in valuation practice. The risk-free rate (R_f) is taken from *A. Damodaran (NYU Stern)* ten-year government bond tables, adjusted using *Moody’s* rating-based default spreads; in the euro area, *German Bunds* anchor R_f , with country spreads applied for sovereign risk differentials. The corporate tax rate (T) is taken from the *OECD Tax Database* and *KPMG Corporate Tax* reports. Sector betas (β) follow *Damodaran’s* published industry betas (unlever/relever as appropriate). The perpetual growth rate (g) for terminal value uses *IMF World Economic Outlook* long-run real GDP projections as a conservative anchor for steady-state growth.

Step 3: Compute the terminal value Beyond the explicit horizon, the Terminal Value (TV) summarizes continuing operations. The standard perpetuity growth (Gordon) model is damodaran2023,koller2020:

$$\text{TV} = \frac{\text{FCF}_n(1+g)}{\text{WACC} - g}, \quad (4)$$

with FCF_n the final-year FCF and g the long-run growth rate.

Step 4: Compute enterprise value Enterprise value is the sum of discounted explicit-period FCFs and discounted TV:

$$\text{EV} = \sum_{i=1}^N \frac{\text{FCF}_i}{(1+\text{WACC})^i} + \frac{\text{TV}}{(1+\text{WACC})^N}. \quad (5)$$

Care is required because DCF is highly sensitive to the WACC and terminal growth assumptions; small perturbations can propagate into large value swings damodaran_{terminal}, alue, koller2020.

3.2 Asset-Based Valuation

Asset-Based Valuation (ABV) estimates firm value from the net book (or adjusted) value of balance-sheet components, emphasizing stocks over flows. Implementation follows Damodaran (2009) and Koller et al. (2020).

Step 1: Identify assets and liabilities Collect, from the most recent balance sheet, major asset classes (intangibles; property, plant and equipment; financial investments; inventories; trade receivables; cash and equivalents) and principal liabilities (long-term debt; provisions; trade payables; short-term borrowings; tax and other obligations).

Step 2: Compute net asset value Compute equity as:

$$\text{Firm Value} = \text{Total Assets} - \text{Total Liabilities.} \quad (6)$$

ABV is most informative in liquidation/distress or where a conservative floor is appropriate damodaran2006, palepu2022; because it down-weights going-concern earning power and inherits historical-cost distortions, practitioners typically use it as a complement rather than a stand-alone estimate penman2013.

3.3 Market Multiples Method

The market-multiples (Comparable Company Analysis, CCA) method infers value from peer ratios, transferring market-implied relationships from a comparable set to the target liu2002equity. The overall workflow below follows Damodaran (2006) and Koller et al. (2020).

Step 1: Select multiples and source data Peers and pre-computed ratios are obtained from the *Re-finitiv Eikon* Peers Valuation tool. The analysis considers P/E, P/CF, P/BV, EV/EBITDA, EV/EBIT, and EV/Sales. To improve robustness, the peer distribution for each multiple is filtered by excluding observations beyond $\pm 2\sigma$ from the mean; the median of the filtered set is used as the transfer statistic, consistent with best practice to limit the influence of outliers liu2002equity, mauboussin2018multiples.

Step 2: Map multiples to target metrics Apply the peer multiple to the target’s most recent, like-for-like financial metric:

$$\text{Estimated Value} = \text{Financial Metric} \times \text{Median Multiple.} \quad (7)$$

Multiple selection is sector-contingent (e.g., P/B in financials; EV/EBITDA in capital-intensive sectors; EV/Sales where earnings are unreliable), with explicit screening for negative denominators and non-recurring items demirakos2004, nissim2012, huikku2020, glaum2006, olbert2024.

3.4 Jump-Diffusion Real-Options Model

The jump-diffusion framework addresses settings in which classical methods underperform—illiquidity, sparse comparables, discontinuities, and managerial timing flexibility—typical of venture investments. The implementation below follows Gibson et al. (2022), rooted in real-options logic with jump processes building on Majd and Pindyck (1987) and estimation techniques for jump components as in Weron et al. (2004).

3.4.1 Model specification and parameters

Firm value V follows a diffusion with compound Poisson jumps:

$$dV = (\mu - \delta)V dt + \sigma V dW + \phi_1 V dq_1 + \phi_2 V dq_2, \quad (8)$$

where μ is the mean growth rate, δ the dividend yield, σ volatility, dW a standard Brownian motion, and dq_1, dq_2 Poisson processes for positive and negative jumps with mean jump sizes ϕ_1, ϕ_2 . Distinct jump channels capture rare, transformative upsides and catastrophic drawdowns gibson2022. Parameters are estimated via a two-step maximum-likelihood calibration on venture data (intensities, sizes, and diffusion terms), consistent with the empirical setting in Gibson et al. (2022).

Calibration and representative estimates Parameters are estimated through a two-step maximum likelihood procedure calibrated on venture-capital data, yielding the following representative values:

$$\begin{aligned} \mu - \delta &= 8.8\% \quad (\text{net growth rate}) \\ \sigma &= 12.16\% \quad (\text{volatility}) \\ \lambda_1 &= 7.55\% \quad (\text{intensity of positive jumps}) \\ \phi_1 &= +55.4\% \quad (\text{mean size of positive jumps}) \\ \lambda_2 &= 4.9\% \quad (\text{intensity of negative jumps}) \\ \phi_2 &= -72.51\% \quad (\text{mean size of negative jumps}) \end{aligned}$$

These estimates inform both the path generation for V_t and the barrier policy, ensuring the simulated dynamics reflect the empirical incidence of rare upside events and catastrophic drawdowns documented in venture outcomes.

3.4.2 Investment rule and Monte Carlo implementation

Let $F(V, K)$ denote the compound real option held by limited partners on committed capital K ; exercise is staged and contingent on crossing a critical threshold V^* proportional to remaining dry powder. As in Gibson et al. (2022), the threshold aligns with a multiple of committed capital (approximately $2.4 \times K$ in their application), translating “invest only when states are favorable” into a barrier rule.

Simulation proceeds over a five-year horizon as follows. For each path of (8), at each step:

1. If $V_t > V^*$, invest a tranche $k \leq K/T$ (with T the horizon); otherwise defer.
2. Update K (remaining commitment) and continue until $t = T$ or exhaustion of K .
3. Record the terminal payoff; discount expected payoffs to obtain $F(V, K)$.

This staged policy embeds the option to wait, expand, or abandon, turning discontinuous upside/downside into valuation and timing guidance gibson2022.

3.4.3 Inputs and data sourcing

The model requires three observable inputs for each startup:

- V_0 : pre-round company valuation,
- K : total committed capital,
- V_{post} : post-round company valuation.

For the empirical application, these are drawn from the *VC Screener* in *Refinitiv Eikon*. Consistent with Gibson et al. (2022), K is interpreted primarily as external LP capital (with marginal GP contribution). If the barrier is hit before full deployment, the option expires out-of-the-money and the payoff is zero; if capital is fully deployed by T , the payoff corresponds to V_T discounted to present value.

The four implementations above deliver a coherent workflow: cash-flow-based intrinsic value (DCF), balance-sheet floors (ABV), market-implied checks (CCA), and an options-theoretic engine for illiquid, jump-prone assets (jump-diffusion). Each carries explicit parameter sourcing and robustness conventions aligned with the cited literature, and each produces estimates that can be compared against realized transactions in subsequent chapters.

4 Performance of DCF, ABV and CCA in M&A and IPO

This chapter moves from methodology to evidence, testing the three canonical valuation approaches on an international set of M&A and IPO deals. The goal is straightforward: confront model-based estimates with realized market values and quantify how close each method gets in practice. The discussion proceeds in three steps. First, we document the sample—its composition, sectoral footprint, and geographic span. Second, we define the accuracy metrics used to compare estimates with benchmarks. Third, we present results with tables, statistical tests, and graphics, drawing out recurrent patterns and method-specific pitfalls.

4.1 Sample Description

The working sample comprises just over 250 companies drawn from the *Orbis BVD* database between 2016 and 2025, spanning more than sixty countries. Advanced economies (United States, United Kingdom, Germany, France, Japan, Australia) sit alongside large emerging markets (Brazil, India, China, South Africa, Turkey, Argentina), allowing heterogeneity in macro conditions, institutional settings, and risk premia to surface in the evaluation.

4.2 Sectoral and Geographic Distribution

Table 1 reports the sectoral split. Machinery, equipment, and recycling account for roughly 18% of observations, with chemicals–plastics slightly above 10%. Categories with smaller representation—such as publishing or primary industries—nonetheless contribute to breadth. Notably, “other services” aggregates over 30% of the sample, reflecting activities that do not map neatly to traditional industry buckets (see Table 1).

Table 1: Sectoral distribution of the sample

Sector	N.
Machinery, equipment, furniture, recycling	46
Chemicals, rubber, plastics, non-metallic products	26
Publishing, printing	18
Primary Sector (agriculture, mining, etc.)	15
Education, Health	9
Gas, Water, Electricity	9
Construction	8
Transport	8
Wholesale & retail trade	10
Banks	1
Food, beverages, tobacco	4
Metals & metal products	4
Hotels & restaurants	5
Wood, cork, paper	5
Post and telecommunications	6
Other services	81

Geographically, Table 2 shows a strong U.S. presence (about 60% of deals), followed by China (6.7%), the United Kingdom (4.3%), Germany and India (3.9% each), and Italy (3.5%). Smaller country shares round out the international scope.

Table 2: Geographical distribution of the sample

Country	N.
United States of America	154
China	17
United Kingdom	11
Germany	10
India	10
Italy	9
Cayman Islands	6
Indonesia	5
Japan	5
Republic of Korea	4
Australia	3
Thailand	3
Other Countries	13

4.3 Evaluation Metrics

Model outputs are benchmarked to observable market references: either the announced value of an M&A transaction or the IPO pricing. The discrepancy between the estimated value and the realized benchmark is the estimation error. The metrics employed are summarized below.

Mean bias captures systematic over- or under-valuation; a value near zero is not sufficient evidence of accuracy, since offsetting positive and negative errors can conceal dispersion. Hence we pair bias with dispersion statistics—MAPE and MdAPE. The median variant is particularly informative in fat-tailed contexts: it dampens outliers and reduces the distortions that arise when y_i is small and percentage errors can explode. Finally, tolerance bands quantify how often estimates land within practically relevant intervals. A $\pm 20\%$ band is a demanding but finance-realistic target; $\pm 50\%$ is looser yet still informative, flagging truly large misvaluations.

Table 3: Evaluation Metrics

Metric	Formula
Mean Bias	$\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$
MAPE (Mean Absolute Percentage Error)	$\frac{100}{n} \sum_{i=1}^n \left \frac{\hat{y}_i - y_i}{y_i} \right $
MdAPE (Median Absolute Percentage Error)	$\text{Median} \left(\left \frac{\hat{y}_i - y_i}{y_i} \right \times 100 \right)$
Tolerance band $\pm 20\%$	% of cases with $\left \frac{\hat{y}_i - y_i}{y_i} \right \leq 0.20$
Tolerance band $\pm 50\%$	% of cases with $\left \frac{\hat{y}_i - y_i}{y_i} \right \leq 0.50$

Where y_i is the market value and $\hat{y}_i$ the model estimate.

4.4 Empirical Results

4.4.1 Bias, MAPE, and MdAPE

Table 4 summarizes aggregate performance across models.

Table 4: Aggregated results of the valuation models

Model	Bias	MAPE	MdAPE
DCF	+12%	38%	25%
ABV	−8%	45%	33%
CCA	+5%	30%	20%

Errors are economically meaningful across the board. MAPEs range roughly from 30% to 45%, indicating average deviations on the order of one-third of observed values. MdAPEs are smaller but still sizable (20–33%), confirming that point estimates rarely coincide with benchmarks.

The DCF displays a modest positive bias (+12%), consistent with terminal-value sensitivity: even slight perturbations in WACC or growth feed disproportionately into enterprise value, yielding optimistic tilts. ABV shows a negative bias (−8%), aligned with its construction as a conservative balance-sheet anchor that ignores going-concern earning power. CCA outperforms on all three indicators: a smaller average bias (+5%) and the lowest MAPE/MdAPE, consistent with its reliance on contemporaneous market information—provided that the peer set is genuinely comparable.

4.4.2 Analysis of the tolerance bands

To complement the aggregates, we use log–log scatter plots comparing realized values (horizontal axis) to model estimates (vertical axis). The 45-degree line marks perfect alignment. Colored tolerance bands highlight absolute percentage deviations: blue for $\pm 20\%$, green for (20–50%), and red for $> 50\%$. The logarithmic scale prevents small and large deals from compressing each other. Figure 1 presents the multiples-based estimates.

The multiples plot shows a visible cluster near the bisector. Numerically, 50 deals ($\approx 20\%$) fall within $\pm 20\%$, and 107 ($\approx 42\%$) within $\pm 50\%$. Almost half the valuations are therefore in an acceptable corridor, with a nontrivial share very close to market values. That said, 98 observations ($\approx 38\%$) breach the 50% threshold, disproportionately among larger transactions. This is consistent with the difficulty of finding truly

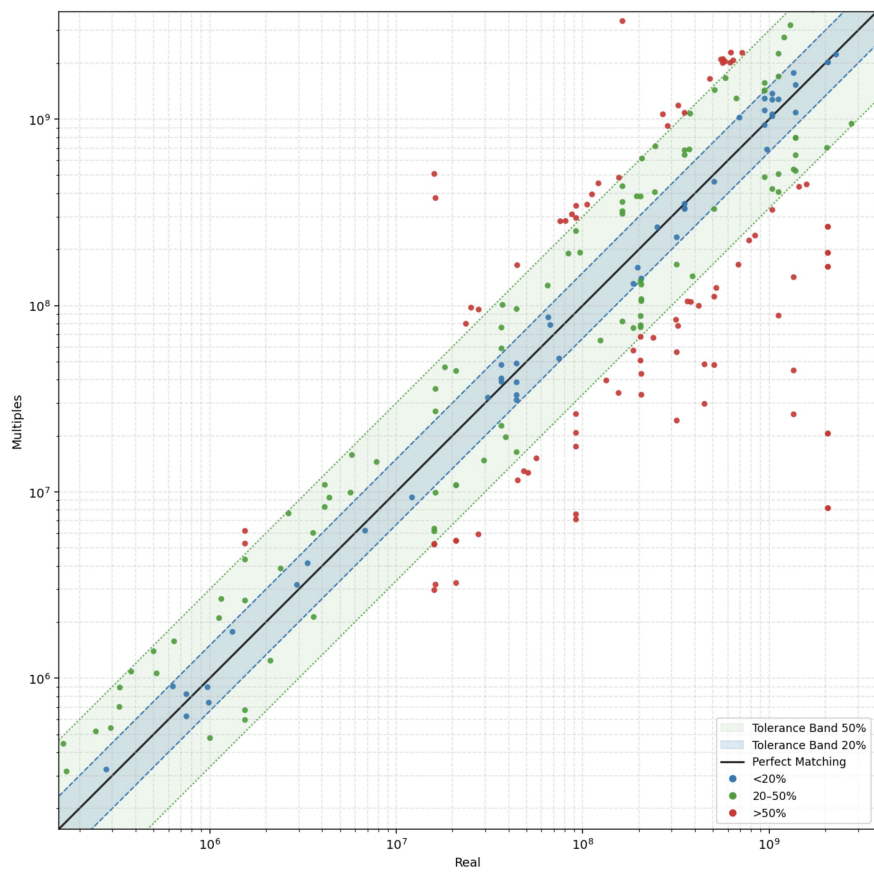


Figure 1: Real vs Multiples

representative peers for very large issuers or firms with distinctive capital or operating structures. Figure 2 turns to ABV.

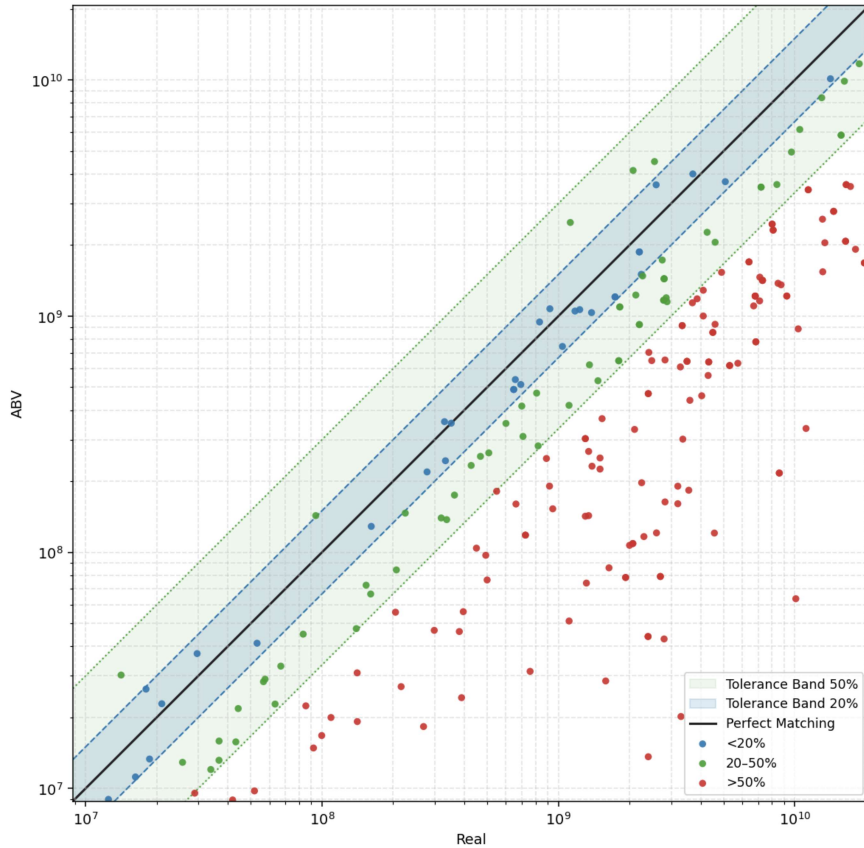


Figure 2: Real vs ABV

Here the under-valuation pattern is clear: points concentrate below the 45-degree line. Only 36 cases ($\approx 14\%$) are within $\pm 20\%$, and 69 ($\approx 27\%$) within $\pm 50\%$. A majority—150 observations (59%)—exceed the 50% error band, in line with the negative bias in Table 4. This is the expected trade-off of a balance-sheet floor applied to heterogeneous, often growth-oriented firms. Finally, Figure 3 reports the DCF.

The DCF scatter is more symmetric around the bisector than ABV's. We count 54 deals ($\approx 21\%$) within $\pm 20\%$ and 94 ($\approx 37\%$) within the 20–50% band. In many instances DCF delivers reasonably close estimates—sometimes rivaling the multiples outcome—yet the presence of pronounced outliers underscores its high variance: small shifts in WACC, growth, or terminal mechanics can produce errors above 100%. In practice, DCF requires careful parameter discipline and triangulation against other methods.

4.4.3 Wilcoxon Test and Heatmap

Beyond visuals, we assess whether differences across multiples are statistically meaningful using a Wilcoxon matched-pairs test on absolute errors. The test compares, deal by deal, which multiple tends to be closer to the benchmark, without imposing distributional assumptions—well-suited to heavy-tailed error structures.

¹All pairwise comparisons were run; the table reports only statistically significant results.

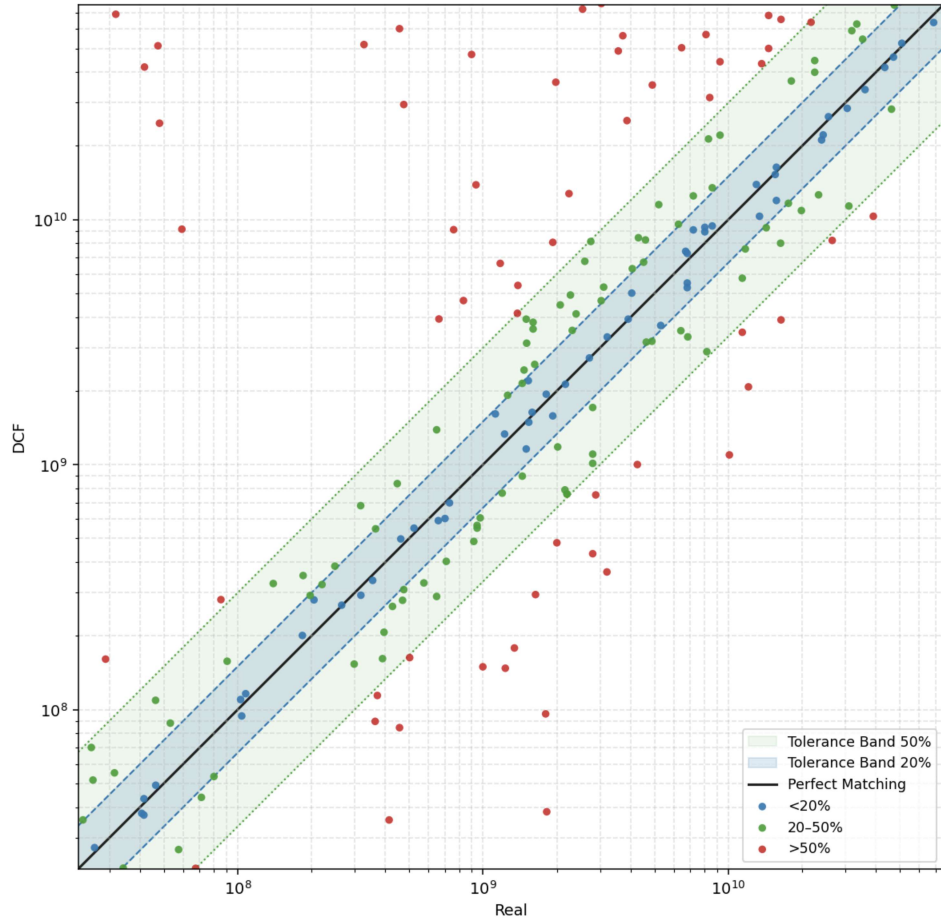


Figure 3: Real vs DCF

Table 5: Wilcoxon Test

Comparing ¹	Better	p-value
EV/EBIT vs EV/Sales	EV/Sales	0.026
EV/EBITDA vs EV/Sales	EV/Sales	0.163
EV/Sales vs P/CF	EV/Sales	0.0432
EV/Sales vs P/E	EV/Sales	0.00618

To complement the test, we report heatmaps of median errors by sector (Figure 4) and by geography (Figure 5). Green = lower median error; red = higher), which make cross-sectional patterns immediately legible. The two tools together—nonparametric testing and error heatmaps—help separate robust tendencies from noise.

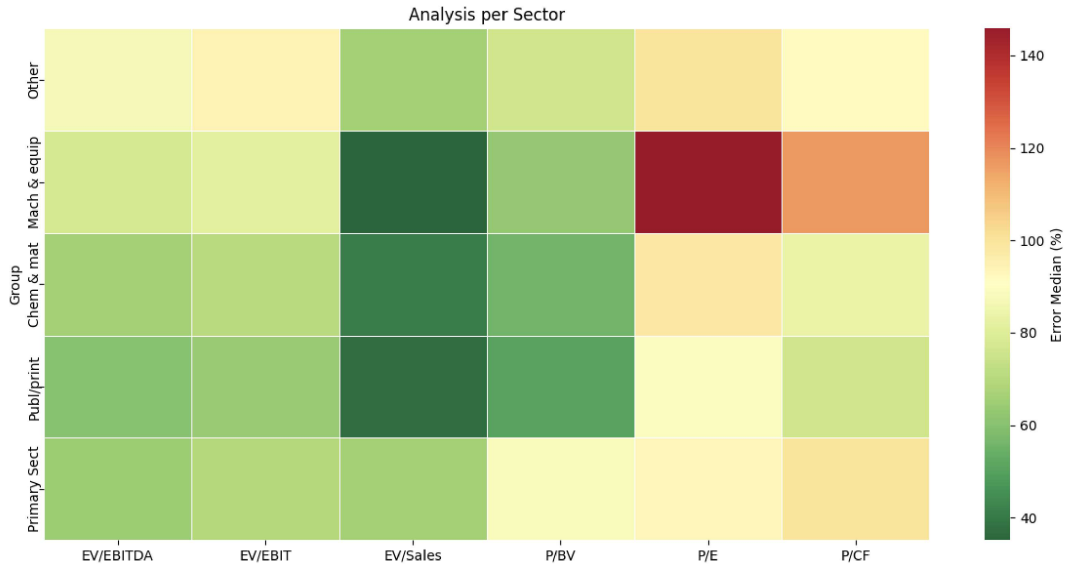


Figure 4: Heatmaps per sector

The takeaways are consistent. No single multiple produces uniformly low errors; median deviations seldom fall below 40–50% even in favorable pockets, underscoring the inherent difficulty of valuation. Within that constraint, EV/Sales emerges as the comparatively most reliable transfer metric. The heatmaps show greener cells for EV/Sales across many sectors and regions, and the Wilcoxon test confirms the edge over EV/EBIT ($p = 0.026$), P/CF ($p = 0.0432$), and especially P/E ($p = 0.00618$), all at or below conventional significance thresholds.

This advantage is most visible in cyclical, capital-intensive segments—Machinery and Equipment, Chemicals and Materials, and the Primary sector—where margins and cash flows are volatile or occasionally negative. In such environments, earnings- or cash-based denominators become unstable, pushing P/E and P/CF toward unusable or noisy territory. Revenues, while cyclical, are more likely to remain positive and comparatively less erratic, making EV/Sales a sturdier anchor when peer comparability is credible.

5 Performance of the Jump Diffusion Model in Startup Valuation

This chapter evaluates the empirical performance of the Jump Diffusion framework when applied to startup valuation. Having benchmarked traditional methods (DCF, ABV, CCA) on M&A and IPO transactions in the previous chapter, we now turn to venture-backed firms—an environment marked by uncertainty, discontinuities, and pronounced heterogeneity. Our objective is to assess whether this option-theoretic approach captures the distinctive features of startups and delivers decision-useful estimates. The discussion proceeds in three parts. First, we document the sample—its sectoral mix, geographic footprint, development stages, and investment-size distribution. Second, we define the accuracy metrics used for model evaluation and analyze aggregate performance, including visual diagnostics. Third, we provide a disaggregated reading

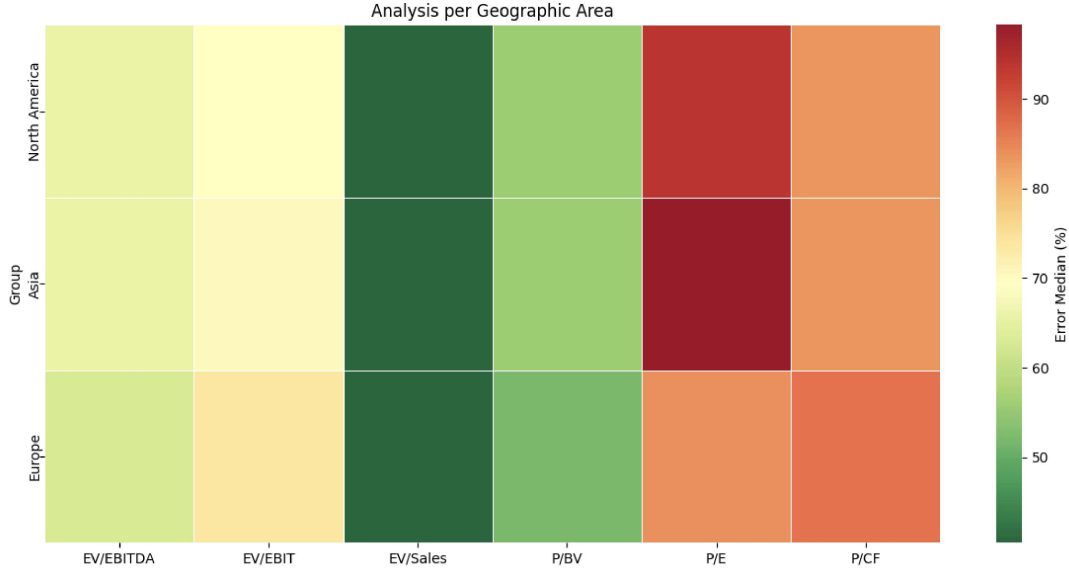


Figure 5: Heatmaps per geographic area

by sector, region, stage, and investment class, highlighting where the model is comparatively reliable and where structural limits emerge.

5.1 Sample Composition

The empirical design leverages more than 3,400 observations from the *Refinitiv Eikon* database, spanning 2015–2025. The sample is deliberately heterogeneous across industry, geography, development stage, and capital intensity (investment size relative to pre-round value). We summarize these dimensions below.

5.1.1 Sector Distribution

Table 6 reports the sectoral split. Technology dominates (over 57%), consistent with secular trends in software, AI, and digital infrastructure. Healthcare and Industrials are also well represented. Consumer-related segments (cyclical and non-cyclical) supply additional breadth. Real Estate, Energy, and Utilities are marginal but useful for heterogeneity.

5.1.2 Geographic Distribution

We classify observations into seven macro-regions. North America is predominant (driven by the United States), reflecting its central role in the global innovation system. Asia and Europe follow, together accounting for roughly one-third of the sample. Latin America, Oceania, Africa, and the Middle East appear with smaller weights but signal the growing dynamism of local ecosystems. An *Other* bucket captures cases with incomplete geography. Table 7 reports the full breakdown.

5.1.3 Development Stage Distribution

We classify deals into Seed, Early Stage, and Expansion at the time of investment.

Table 6: Sectoral distribution of the dataset

Sector	Observations
Technology	2000
Healthcare	453
Industrials	315
Consumer Cyclicals	314
Consumer Non-Cyclicals	149
Financials	128
Academic & Educational Services	34
Basic Materials	33
Real Estate	31
Utilities	17
Energy	11
Institutions, Associations & Organizations	4
Government Activity	3

Table 7: Geographical distribution of the dataset

Region	Observations
North America	1773
Asia	860
Europe	506
Latin America	100
Oceania	67
Africa	38
Middle East	27
Other	145

Table 8: Investment stage composition

Stage	Observations
Seed	340
Early Stage	1682
Expansion	1470

Seed investments are the least represented and the riskiest (concept/prototype). Early Stage concentrates the plurality of observations (product finalized, early customers). Expansion targets scale-up and market consolidation and is the second-largest group.

5.1.4 Investment Size Distribution (K/V_0 Ratio)

Capital intensity is proxied by K/V_0 , i.e., invested capital relative to pre-round valuation. Table 9 reports the five-class partition used throughout.

Table 9: Investment size distribution

Category	Observations
Very Small ($K/V_0 < 25\%$)	2112
Small ($25\% \leq K/V_0 < 35\%$)	457
Medium ($35\% \leq K/V_0 < 50\%$)	282
Large ($50\% \leq K/V_0 < 75\%$)	245
Very Large ($K/V_0 \geq 75\%$)	396

This partition is operationally meaningful because the jump-diffusion policy compares simulated firm value against a barrier linked to K . The ratio K/V_0 therefore governs the likelihood of exercise and the distribution of payoffs.

5.2 Overall Accuracy of the Model

To evaluate predictive performance, we follow the metrics introduced in Section 4.3 and add complementary indicators (MAE, RMSE, sMAPE, MedAE, R^2). Table 10 reports formulas and values.

Table 10: Overall accuracy metrics for the Jump Diffusion model

Metric	Formula	Value
MAE (Mean Absolute Error)	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	1.676×10^8
RMSE (Root Mean Squared Error)	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	7.441×10^8
MAPE (Mean Absolute Percentage Error)	$\frac{100}{n} \sum_{i=1}^n \left \frac{y_i - \hat{y}_i}{y_i} \right $	32.07 %
sMAPE (Symmetric MAPE)	$\frac{100}{n} \sum_{i=1}^n \frac{ y_i - \hat{y}_i }{\frac{ y_i + \hat{y}_i }{2}}$	44.02 %
MedAE (Median Absolute Error)	$\text{median}(y_i - \hat{y}_i)$	1.005×10^7
R^2 (Coefficient of Determination)	$1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$	85.65 %

Where y_i represents the market value, $\hat{y}_i$ is the model value, and $\bar{y}$ is the mean of the market values.

5.2.1 Interpretation of the global metrics

The coefficient of determination ($R^2 = 0.8565$) indicates that the model explains more than 85% of cross-sectional variation in market values—suggestive of a strong fit. Read in isolation, however, R^2 can be misleading: it does not guarantee low absolute errors and is insensitive to tail behavior. We therefore consider level-based errors. The MAE is 1.676×10^8 and the RMSE is 7.441×10^8 , with the latter over four times the former—evidence of large outliers. Consistently, the median absolute error (MedAE) is only 1.005×10^7 , far below the MAE, underscoring skew from a small set of extreme deviations.

Given the wide scale of valuations (from sub-million to multi-billion), absolute errors are complemented by percentage metrics. The MAPE is about 32%, consistent with the uncertainty inherent in startup valuation. sMAPE is higher (44%), signaling cases where the model materially underestimates firm value—an asymmetry that will reappear in the visual diagnostics.

5.2.2 Scatterplot of Real vs Predicted Values

We visualize performance with a log-log scatterplot of observed values y (horizontal) versus predictions $\hat{y}$ (vertical). The 45-degree dashed line represents perfect alignment. Each point is color-coded by absolute percentage error (APE), ranging from blue (low error) to red (high error).

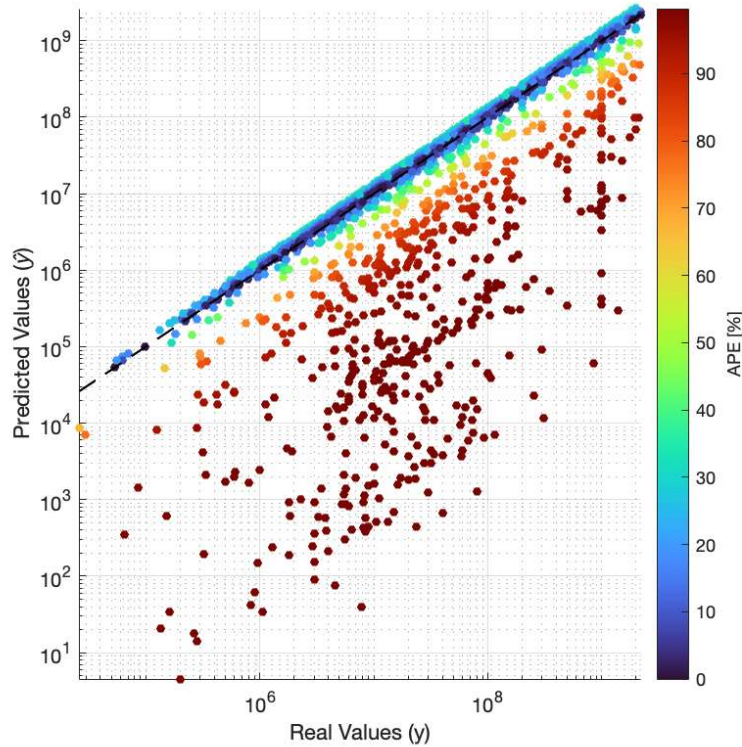


Figure 6: Real vs Predicted

Across most of the support, accurate and inaccurate predictions coexist: at a given y , some observations lie close to the bisector while others diverge substantially. The cloud's slight displacement below the 45-degree line indicates a systematic underestimation bias, consistent with the sMAPE finding. In short, while average fit is strong, reliability at the individual-deal level is uneven, motivating a deeper cut.

5.2.3 Distribution of errors by class

Following the convention attributed to lewis1982, we classify percentage errors as *Excellent* ($\leq 10\%$), *Good* (10–20%), *Moderate* (20–50%), and *Poor* ($>50\%$). Table 11 reports frequencies.

Table 11: Errors Classification

Class	Observations	Percentage (%)
Excellent ($\leq 10\%$)	644	18.32
Good (10–20%)	807	22.95
Moderate (20–50%)	1443	41.04
Poor ($> 50\%$)	622	17.69

Roughly 41% of forecasts fall within 20% of the benchmark (*Excellent*+*Good*); the modal class is *Moderate* (41%), implying many estimates deviate by up to half of the observed value. A meaningful minority (17.7%) exceeds 50% error. This distribution reflects the intrinsic difficulty of valuing early-stage firms. Still, over 80% of predictions land within a 50% band—an improvement relative to the performance of traditional methods on mature companies—supporting the relevance of the jump-diffusion approach in this domain.

5.3 Disaggregated Performance Analysis

We now segment performance by sector, geography, stage, and investment size to identify where the model is most reliable and where it struggles. These subgroup views surface patterns obscured in aggregates.

5.3.1 Differences by sector

Figure 7 reports boxplots for the main sectors (panel a) and for smaller sectors (panel b). Healthcare (H) and Energy (E) display large interquartile ranges (IQRs), indicating high central dispersion rather than isolated tail events; Healthcare exhibits a median near zero (low directional bias with high uncertainty), while Energy combines wide dispersion with a negative median (systematic underestimation). Real Estate (RE), Utilities (U), Basic Materials (BM), Industrials (I), Consumer (CNC, CC), and Financials (F) show tighter IQRs but frequent outliers—most observations are stable, punctuated by episodic failures.

5.3.2 Differences by geographical area

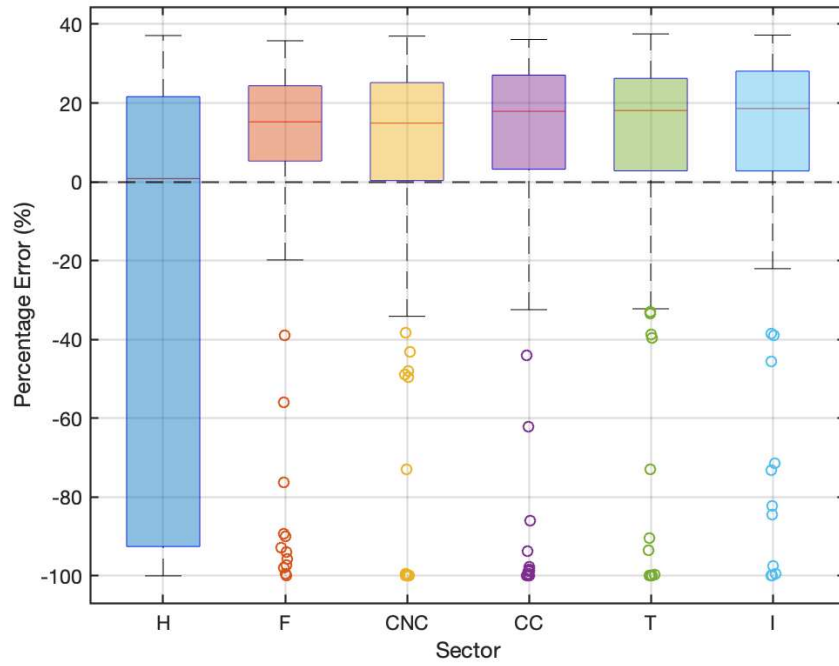
Figure 8 plots error distributions by region (panel a: larger regions; panel b: smaller regions). North America (NA) shows wide boxes (dispersion concentrated in the center, few extreme tails). Europe (E) and Asia (As) are more compact in the central 50% but feature more frequent outliers—typically consistent predictions punctuated by large misses.

5.3.3 Differences by development phase

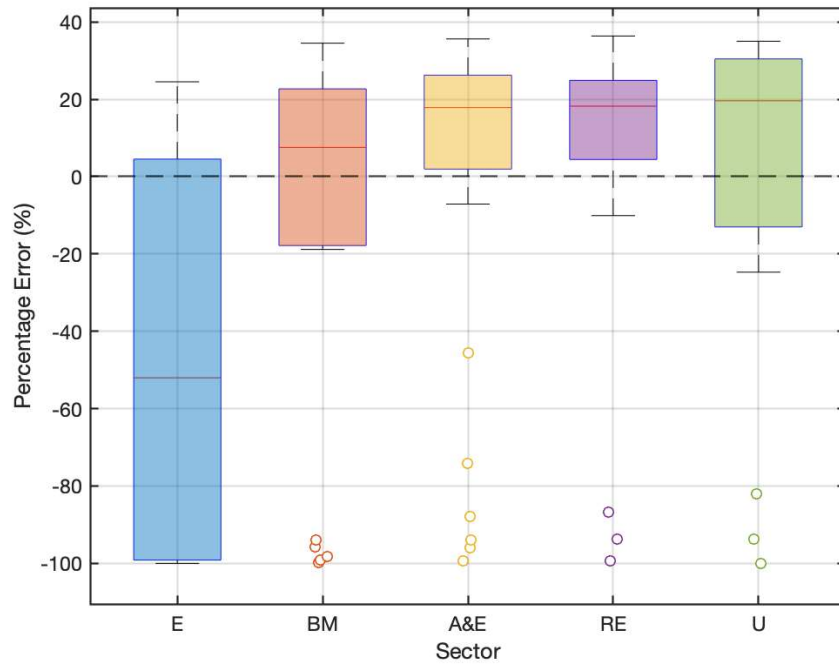
Seed, Early, and Expansion phases display broadly similar IQRs and outlier counts (Figure 9). Contrary to the expectation that maturity would stabilize estimates, dispersion and tail risk remain comparable across stages, suggesting that model variance is not materially attenuated by phase.

5.3.4 Investment Size as discriminating factor

By contrast, investment size K/V_0 is decisively discriminating (Figure 10). We observe a well-defined window (approximately 25–35%) in which bias is minimal and dispersion compresses markedly—estimates are balanced and

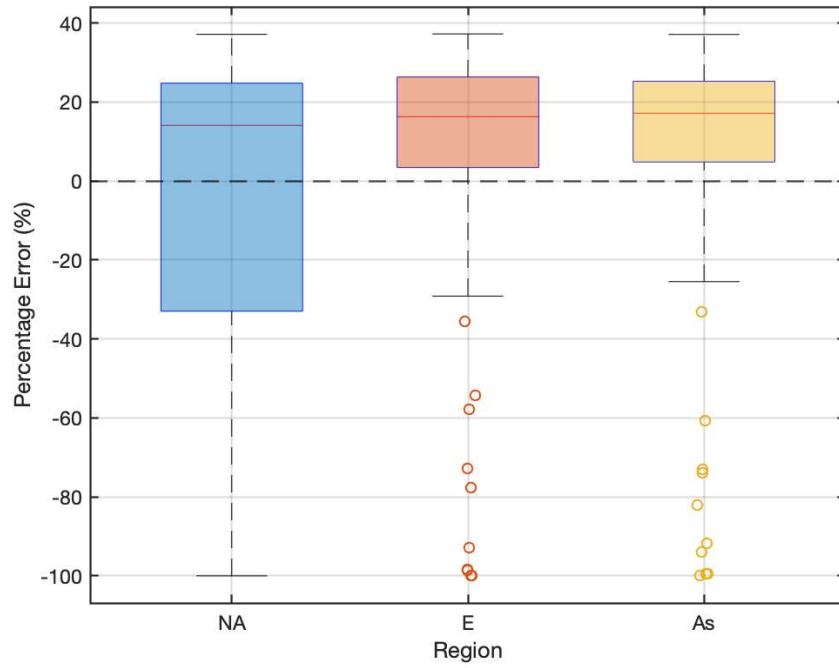


(a) Most representative sectors

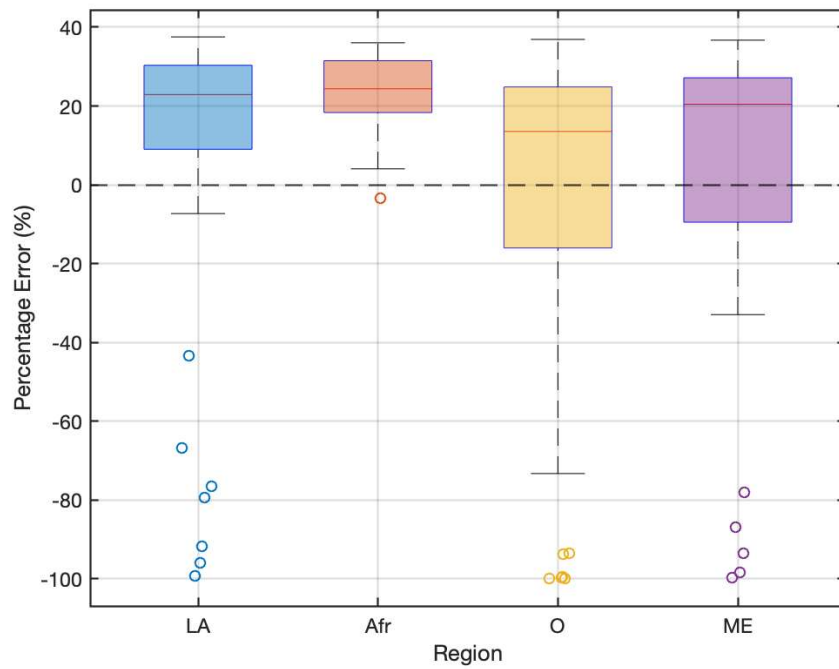


(b) Less representative sectors

Figure 7: Boxplots of distribution error per sector



(a) Most representative geographic areas



(b) Less representative geographic areas

Figure 8: Boxplots of distribution error per geographic areas

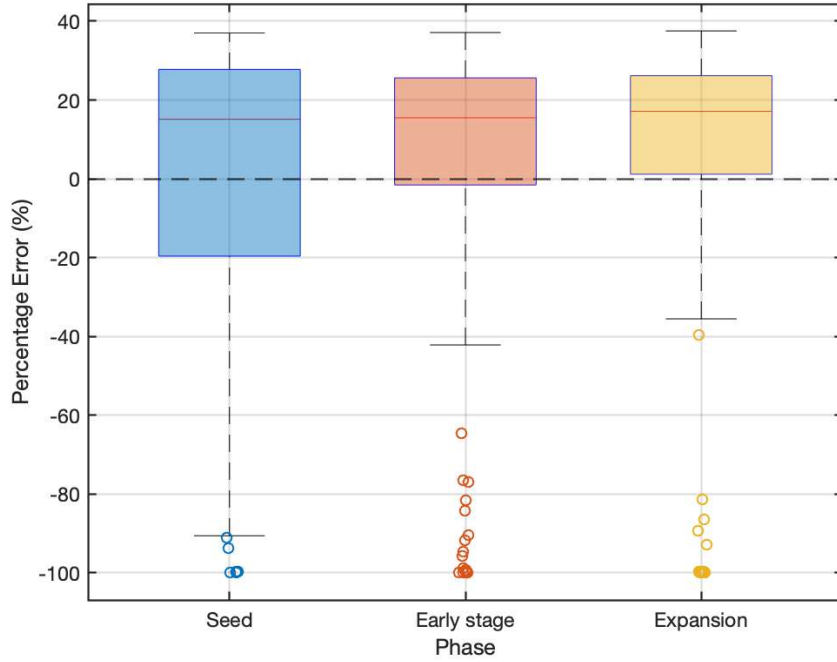


Figure 9: Error distribution per phase

consistent. Performance deteriorates outside this interval. For very small K/V_0 , mild overestimation appears; for very large K/V_0 , the model often produces zero valuations. This is not a numerical artifact but a structural implication of the barrier policy: when K is too large relative to V_0 , the probability that simulated paths cross the investment threshold becomes negligible, so payoffs collapse to zero. When compared with positive realized values, percentage errors approach -100% , reflecting the model's design rather than a conventional miss.

5.3.5 Overall

In sum, while sector and region modulate error magnitudes, they do not eliminate tail risk; phase effects are limited. Investment size is the primary determinant of reliability. Three findings follow. First, sizeable misvaluations and extreme errors persist across subgroups. Second, some contexts (e.g., traditional sectors or established regions) appear somewhat more stable, but the evidence is partial. Third, the 25–35% K/V_0 range constitutes a *best-case* operating region for the jump-diffusion implementation: within it, estimates are comparatively robust and balanced; outside it—especially at very high K/V_0 —the structural barrier logic renders outputs unreliable, calling for alternative specifications or complementary methods.

6 Conclusions

The evidence gathered in this thesis leads to a clear takeaway: no single valuation framework fully captures the complexity of the firms examined, nor is any approach universally applicable. Nevertheless, the empirical tests yield informative results that clarify where each method contributes most. Table 12 synthesizes these findings by mapping models to use-cases and pointing to the supporting exhibits. Given the nontrivial error levels documented throughout, these conclusions should be read with appropriate caution.

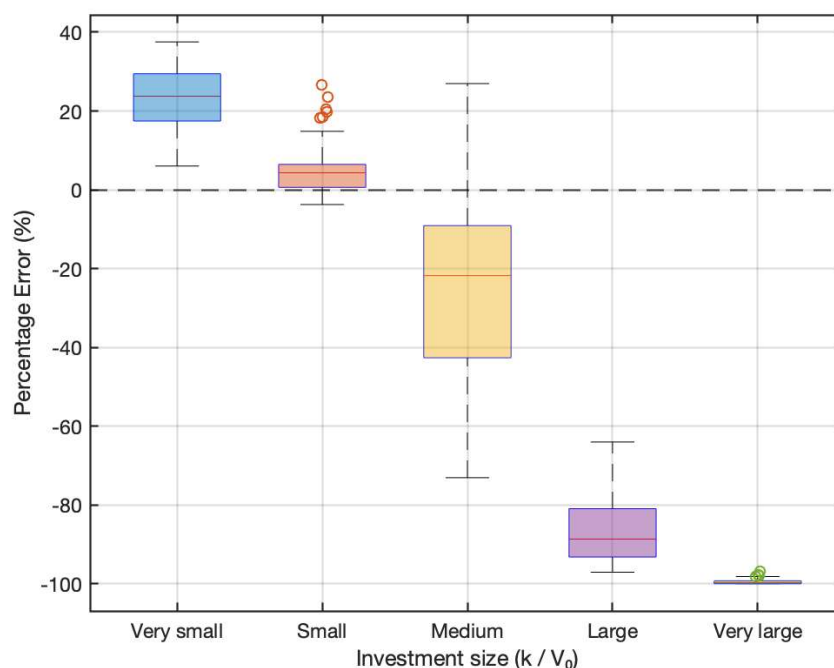


Figure 10: Investment Size Errors

Table 12: Summary by method

Method	Better Application	Evidence
ABV	Not excelling in any specific context	Ch. 3, Fig. 2
DCF	Heterogeneous - vulnerable to parameter sensitivity	Ch. 3, Fig. 3
Multiples (CCA)	EV/Sales in Machinery & Equipment, Chemicals/Materials, and Primary sector	Ch. 3, Tab. 5, Fig. 4 and 5
Jump Diffusion	Startups with $K/V_0 \approx 25-35\%$	Ch. 4, Fig. 10

First, the results confirm that ABV is unsuitable as a stand-alone valuation device; at most, it provides a conservative floor beneath which a counterparty is justified in rejecting a transaction. Second, for DCF and market multiples,

we do not find dominance of one over the other. Both can be close to realized values in some settings and materially off in others. The DCF approach embeds firm-specific economics (FCF construction, WACC, terminal mechanics), yet it is vulnerable to parameter sensitivity: modest shifts in growth or discount rates propagate disproportionately to value. It also remains blind to qualitative drivers not visible in accounting data—organizational execution, geopolitical exposure, governance frictions—which can alter market pricing. Moreover, recent losses in statements may be informative in different directions: if investment-driven, DCF can understate value; if distress-driven, it can overstate it. On the other hand, multiples provide a transparent market anchor and a relative positioning versus peers, but they can suffer from circularity and survivorship bias. When the peer set itself is influenced by valuations that reflect the target’s narrative, self-referential errors arise and reliability deteriorates. These limits indicate that purely quantitative procedures are insufficient. In M&A and IPO settings, transaction prices embed qualitative assessments—management quality, strategic alliances, execution capacity, reputational capital, and ESG posture. Their exclusion helps explain the observed misalignments between model and market and, at the same time, points to practical improvements. Two operational enhancements follow naturally. First, differentiated discount rates that incorporate qualitative risk diagnostics can better reflect firm-specific exposures. Second, structured scenario analysis (base, optimistic, pessimistic) can replace single-point estimates with ranges that are consistent with plausible paths. A further avenue is a hybrid construct that bridges DCF and multiples by deriving *justified* multiples from fundamentals—ROIC, growth, and WACC—rather than imposing industry aggregates. In doing so, the relative metric inherits the firm’s economic engine, reducing dispersion and separating firms with similar growth but different profitability.

Regarding startups, traditional methods are inapplicable given the lack of histories and comparables. In our tests, the Jump Diffusion framework is the most promising alternative, but it also exposes a critical edge case: when K is very large relative to V_0 , simulated paths rarely clear the investment barrier and the model collapses to zero payoffs. This is unrealistic in contexts where very large rounds fund genuinely transformative technologies. To address this weakness, quantitative engines should be complemented with structured qualitative scoring. Methods such as the Berkus approach—despite its subjectivity—force explicit evaluation of idea quality, team, prototype readiness, relationships, and initial traction. Likewise, the Risk Factor Summation Method introduces disciplined adjustments for managerial, technological, regulatory, and go-to-market risks. While imperfect, these overlays capture dimensions that the Jump Diffusion process omits.

In conclusion, the task is not to select a single, “perfect” model (and our evidence shows that such a tool does not exist). The path forward is an integrated framework that blends quantitative rigor with qualitative assessment, and that flexes method choice to context. Established models—DCF, CCA, ABV, and real-options-based Jump Diffusion—remain essential starting points; combined thoughtfully, enhanced with scenario ranges and justified multiples, and augmented by qualitative diagnostics, they can better reflect the complex reality of valuation and guide both research and practice toward more reliable decisions.

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A Appendix

A.1 Implementation of the FCF in the DCF Model

A.1.1 Projection Methodology

In order to project revenues, costs, and cash flows, historical averages of growth rates and margins are computed and then applied to future values. The general formulation for a ratio-based projection is:

$$g_X = \frac{1}{n} \sum_{t=1}^n \frac{X_t}{Y_t}, \quad X_{t+1} = Y_{t+1} \cdot g_X \quad (9)$$

where g_X represents the average historical ratio between variable X and reference variable Y .

A.1.2 Revenue, Costs, and EBITDA

$$g_R = \frac{1}{n} \sum_{t=1}^n \frac{R_t - R_{t-1}}{R_{t-1}}, \quad R_{t+1} = R_t \cdot (1 + g_R) \quad (10)$$

$$g_{COGS} = \frac{1}{n} \sum_{t=1}^n \frac{COGS_t}{R_t}, \quad COGS_{t+1} = R_{t+1} \cdot g_{COGS} \quad (11)$$

$$g_{EBITDA} = \frac{1}{n} \sum_{t=1}^n \frac{EBITDA_t}{R_t}, \quad EBITDA_{t+1} = R_{t+1} \cdot g_{EBITDA} \quad (12)$$

A.1.3 CapEx and D&A

$$CapEx_t = PPE_t^{end} - PPE_t^{beg} + D\&A_t \quad (13)$$

$$g_{CapEx} = \frac{1}{n} \sum_{t=1}^n \frac{CapEx_t}{PPE_t^{beg}}, \quad CapEx_{t+1} = PPE_{t+1}^{beg} \cdot g_{CapEx} \quad (14)$$

$$g_{D\&A} = \frac{1}{n} \sum_{t=1}^n \frac{D\&A_t}{PPE_t^{beg}}, \quad D\&A_{t+1} = PPE_{t+1}^{beg} \cdot g_{D\&A} \quad (15)$$

A.1.4 NOPAT

$$EBIT_{t+1} = EBITDA_{t+1} - D\&A_{t+1} \quad (16)$$

$$Taxes_{t+1} = EBIT_{t+1} \cdot T \quad (17)$$

$$NOPAT_{t+1} = EBIT_{t+1} - Taxes_{t+1} \quad (18)$$

A.1.5 Net Working Capital

Efficiency ratios are computed as:

$$DSO = \frac{1}{n} \sum_{t=1}^n \frac{AR_t}{R_t} \cdot 360, \quad AR_{t+1} = \frac{DSO \cdot R_{t+1}}{360} \quad (19)$$

$$DIO = \frac{1}{n} \sum_{t=1}^n \frac{Inv_t}{COGS_t} \cdot 360, \quad Inv_{t+1} = \frac{DIO \cdot COGS_{t+1}}{360} \quad (20)$$

$$DPO = \frac{1}{n} \sum_{t=1}^n \frac{AP_t}{COGS_t} \cdot 360, \quad AP_{t+1} = \frac{DPO \cdot COGS_{t+1}}{360} \quad (21)$$

Other working capital items are scaled to revenues:

$$X_{t+1} = R_{t+1} \cdot g_X, \quad X \in \{OtherAssets, OtherLiabilities, AccruedSalaries\} \quad (22)$$

Finally:

$$\begin{aligned} NWC_{t+1} = & (AR_{t+1} + Inv_{t+1} + OtherAssets_{t+1}) \\ & - (AP_{t+1} + AccruedSalaries_{t+1} + OtherLiabilities_{t+1}) \end{aligned} \quad (23)$$

A.1.6 Free Cash Flow

$$FCF_{t+1} = NOPAT_{t+1} + D\&A_{t+1} - CapEx_{t+1} - \Delta NWC_{t+1} \quad (24)$$

A.2 Variable Description

Symbol	Description
R_t	Revenue at time t
$COGS_t$	Cost of Goods Sold
$EBITDA_t$	Earnings before interest, taxes, depreciation, amortization
$CapEx_t$	Capital Expenditures
PPE^{beg}, PPE^{end}	Property, Plant & Equipment at beginning / end of period
$D\&A_t$	Depreciation & Amortization
$EBIT_t$	Earnings before interest and taxes
T	Corporate tax rate
$NOPAT_t$	Net Operating Profit After Taxes
AR_t, Inv_t, AP_t	Accounts Receivable, Inventory, Accounts Payable
DSO, DIO, DPO	Days Sales Outstanding, Inventory Outstanding, Payables Outstanding
NWC_t	Net Working Capital